Introduction to Python

Objective

In this module, you will get a basic understanding of python programming, Virtual environment, Package manager, version differences of python programming and detailed knowledge about the python installation and environment setup for working with python.

Topic

- Python programming history & features
- Python compiler and IDE installation
- Virtual Environment
- Pip – Package Manager

Hands-on:

- How to set up an environment for python development.

Basics of Python

Objective

In this module, you will get a basic understanding of python Syntax and a detailed understanding of Input/Output [I/O] operations, Variables, Operators Datatypes and Data structure.

Basics of Python

Topic

- Python Syntax Overview, Indentation, comments
- Variable declaration
- Datatypes and data structure
  - * Primitive
  - * Non-primitive
- Operators in python

Hands-on:

- How to write a basicPython code, variabledeclaration datatypes, and operators usage.

Program flow/ Data flow of Python

Objective

In this module,you will get a detailedunderstanding of conditional statements, looping, control statements of python.

Topics

- Conditional Statements
  - * if statement
  - * if … else statement
  - * if … elif… else statement
- Looping
  - * for loop
  - * for with else statement
  - * while loop
  - * while with else statement
- Control Statements

  break

  Continue

  pass
- Assert Statement

Program flow/ Data flow of Python

Hands-on:

- How to use conditional, looping and controlstatements in python.

Function in Python

Objective

In this module, you will get a detailed understanding of writing function, the scope of variables, function with arguments, keywordarguments, lambda functionsand Modules in python.

Topics

- Syntax of Function
- Function with *args & **kwargs
- Scope of variables
- Lambda function with map, filter,reduce method
- DocString
- Modules and standardModules

Hands-on

How to create and use functionsand different types of argument, lambda function, and modules in python.

File Handling in Python

Objective

In this module,you will get a detailedunderstanding of file concepts like create, read, write, update, delete operation of file with python.

Topics

- File Opening modes
- Context Manager in python
- File Operations
    * Open
    * Create
    * Read
    * Write
    * Update
    * Delete

Hands-on

How to create and use file operation in python.

Exception Handlingin Python:

Objective

In this module,you will get a detailedunderstanding of exceptionhandling and creating your own exception classes in python.

Exception Handlingin Python:

Topics

- Types of Errorsin python
- Exception handling with
  - * try ... except
  - * try ... except...finally
  - * try ... except...else
- Multiple Exception
- Raising Exception
- User-defined Exception

Hands-on

How to create and use built-inand user-defined exception handling in python.

Oops in Python

Objective

In this module,you will get a detailedunderstanding of Object-Oriented Programming Concepts in python.

Topics

- Oops Concepts with programming syntax
  - * Class
  - * Object
  - * Polymorphism

Topics

- * Encapsulation
- * Inheritance
- Types of Methodsin python

Oops in Python

Hands-on

How to use Object-oriented programming concepts in python.

Core Concepts in Python

Objective

In this module,you will get a detailedunderstanding of iterators, generators, decorators, in python.

Topics

- Iterator
- Generator
- Decorator

Hands-on

- How to use core concepts and application of core conceptsin python.

Comprehension in Python

Objective

In this module,you will get a detailedunderstanding of List, Dictionary comprehension and specialized sorts in python.

Comprehension in Python

Topics

- Comprehensions
    * List
    * Nested List
        * if statement
        * if ... else statement
        * Nested if ... else statement
- Dictionary
- Sorting
    * List
    * Dictionary

Hands-on

How to use comprehensions and sorting in python.

Thread and DateTime in Python

Objective

In this module,you will get a detailedunderstanding of multi-threading concepts, datetime module alone with sleep and execution of code in python.

Topics

- Terms in threading
- process
- thread
- multithreading
- Time complexity
- Thread Life cycle
- Programming with Threading & Multithreading
- Synchronization
- Sleep and execution time of code
- DateTime module

Comprehension in Python

Topics

- Comprehensions
    * List
    * Nested List
        * if statement
        * if ... else statement
        * Nested if ... else statement
- Dictionary
- Sorting
    * List
    * Dictionary

Hands-on

How to use comprehensions and sorting in python.

Thread and DateTime in Python

Objective

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Topics

- Terms in threading
- process
- thread
- multithreading
- Time complexity
- Thread Life cycle
- Programming with Threading & Multithreading
- Synchronization
- Sleep and execution time of code
- DateTime module

Thread and DateTime in Python

Hands-on

How to use threading and data time concepts in python.

MySQL with Python

Objective

In this module,you will get a detailedunderstanding of SQL statements and database connection along with CRUD operation using python.

Topics

- SQL statements & Operations
  - Create
  - Read
  - Update
  - Delete
- Python – SQL connector packageinstallation
- Python with CRUD Operations
- Commit & Rollback
- SQL Related Exception Handling

Hands-on

How to use and manipulate data in a database usingpython.

Network programming with Python

Objective

In this module,you will get a detailedunderstanding of Networkprogramming, Client & server concepts with python.

Topics

- Terms and Basicsof network programming
- The architecture of data transmission between sender and receiver using python
- Getting data from the remoteserver
- Client & Server-side programming

Hands-on

How to use and networking module of python, data transmission between client to server and server to client python.

Regular Expression with Python

Objective

In this module, you will get a detailed understanding of writing Regex with python.

Topics

- Regex Syntax
- Quantifiers
- Metacharacters
- Special Sequences

Sets

- Python re module
- Methods with regexusage

Regular Expression with Python

Hands-on

How to use and write regex in python.

GUI programming with Python

Objective

In this module,you will get a detailedunderstanding of developing the GUI application using the PyQt5 module with python.

Topics

- Introduction
- Components and Events
- An Example GUI
- Widgets
- Layout Management
- Signals & Slots
- QMessagesBox, QDialog
- Database Handling

Hands-on

How to develop a GUI application with PyQt5 and python.

APIaccess with Python

Objective

In this module,you will get a detailedunderstanding of accessingopen APIs using python.

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APIaccess with Python

Topics

- Google Text to Speech
- Google Speech to Text
- OpenWeatherMap

Hands-on

How to use open APIs using python.

DataScience with Python

Objective

In this module, you will geta Basic understanding of data sciencemodules in python.

Topics

- Pandas – Series and Dataframe
- Numpy
- Matplotlib

Hands-on

- How to use data science modulesof python.

The project with Python

Objective

In this module, you are goingto develop an application for own scenario.

Topics

- Creating own application with any one of the frameworks
- Django App
- PyQt5 App
- Console orientedCore app